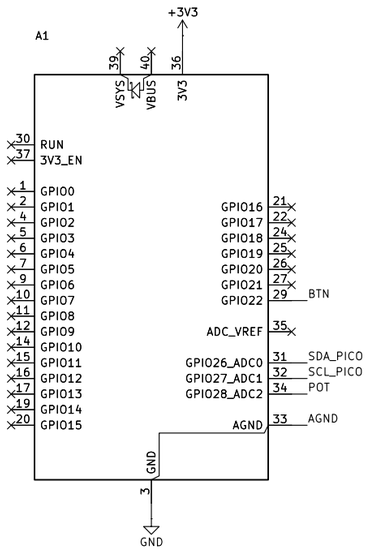
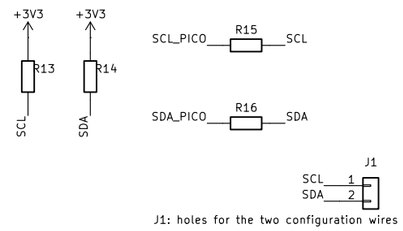
## 1. Microcontroller: Raspberry Pi Pico (socketed)

I2C1: SDA = GP26, SCL = GP27 (Wire1).  Button = GP22.
Potentiometer = GP28 (ADC2).  Build the firmware with FRUTAMIDI_PCB.



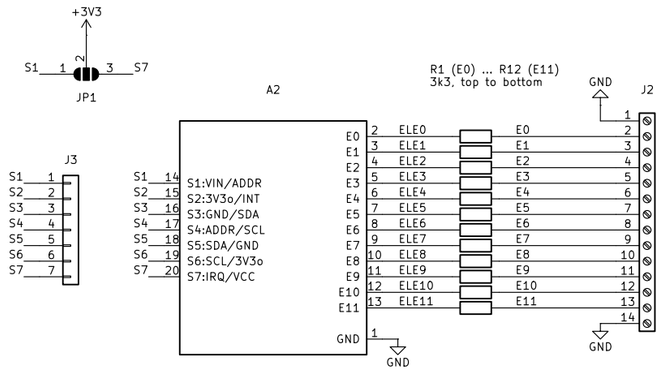

## 2. I2C bus

R13/R14: 4.7k pull-ups to 3V3 (rise time at 400 kHz, T-050).
R15/R16: 220R between the Pico pins and the bus (wrong-wire current limit, T-055).




J1: holes for the two configuration wires

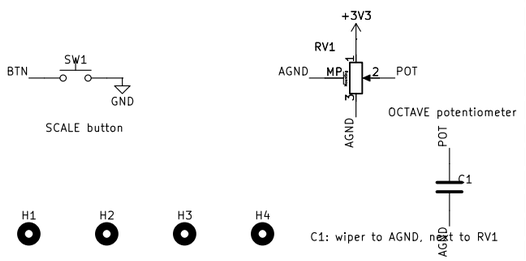
## 3. Touch sensing: MPR121 module, electrode resistors, screw terminals




Module configuration (component side):
  Original module (no STEMMA QT):  JP1 bridge 'ORIG' (3V3 -> S1)   wires SDA -> S5   SCL -> S6
  STEMMA QT module (current):      JP1 bridge 'QT'   (3V3 -> S7)   wires SDA -> S3   SCL -> S4
Module GND enters through the electrode row (pin 1), the same on both revisions.

R1-R12: 3.3k in series with each electrode (ESD current limit, T-050);
pulse-rated DIN0414, lying flat (flashover gap, T-060).

## 4. User controls and mounting holes




Raspberry Pi Pico + Adafruit MPR121 (1982), single-sided PCB
**github.com/catuta0/fruta-midi-controller**

Sheet: /
File: frutamidi.kicad_sch

***Title: FrutaMIDI — capacitive MIDI controller***

| Size: A3 | Date: 2026-11-07 | **Rev: 1.2** |
|---|---|---|
| KiCad E.D.A. 10.0.4 | | Id: 1/1 |